

KRESHCHENSKOYE, Russian Federation

WMO: 295240

Lat: 55.8593N

Lon: 80.0608E

Elev: 422

StdP: 14.47

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.0	(c) -28.8	(d) -39.8	(e) 0.6	(f) -34.3	(g) -33.9	(h) 0.8	(i) -28.5	(j) 17.5	(k) 13.7	(l) 15.4	(m) 16.2	(n) 2.0	(o) 250	(p) 0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	83.2	66.5	80.3	65.5	77.5	64.3	69.6	79.1	67.8	77.2	66.0	74.5	6.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.9	74.4	64.3	91.6	72.4	62.6	86.3	70.9	34.0	78.8	32.5	77.2	31.1	74.5	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.3	(b) 14.2	(c) 12.6	(e) -42.8	(f) 88.4	(g) 6.6	(h) 4.5	(i) -47.6	(j) 91.6	(k) -51.5	(l) 94.3	(m) -55.2	(n) 96.8	(o) -60.1	(p) 100.2		
			DB			4.5										
			WB			6.4										

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	-1.7	3.9	18.8	36.8	51.2	60.8	64.2	58.9	48.3	36.1	17.8	4.4	
	DBStd	25.42	14.99	13.41	11.50	10.54	9.65	7.47	5.77	6.48	7.72	8.97	14.18	15.55	
	HDD50	7340	1603	1291	966	412	108	11	0	8	123	438	966	1414	
	HDD65	11642	2068	1711	1431	847	436	161	85	208	504	896	1416	1879	
	CDD50	1297	0	0	0	15	145	336	441	282	71	7	0	0	
	CDD65	124	0	0	0	0	8	36	61	18	1	0	0	0	
	CDH74	1492	0	0	0	2	172	479	597	215	27	0	0	0	
	CDH80	280	0	0	0	0	35	95	119	30	1	0	0	0	
Wind	WSAvg	5.8	5.6	5.9	6.6	6.9	6.7	5.0	4.5	4.5	5.3	6.0	6.4	6.2	
Precipitation	PrecAvg	17.2	0.9	0.7	0.7	0.9	1.3	2.2	2.6	2.4	1.4	1.7	1.4	1.2	
	PrecMax	22.9	1.7	1.1	1.9	2.6	3.4	4.5	6.3	4.6	3.7	3.6	2.7	2.4	
	PrecMin	11.3	0.1	0.0	0.1	0.2	0.3	0.1	0.2	0.5	0.3	0.7	0.3	0.3	
	PrecStd	3.0	0.4	0.2	0.4	0.6	0.7	1.3	1.3	0.9	0.8	0.8	0.5	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	34.0	44.5	70.8	84.2	86.0	87.3	83.8	77.9	66.5	44.9	34.1	
		MCWB	30.8	31.9	37.6	54.6	61.6	67.6	70.4	68.4	62.1	51.6	41.3	32.3	
	2%	DB	27.8	30.5	40.1	64.3	78.7	82.9	83.1	79.7	71.8	59.7	40.8	31.9	
		MCWB	26.5	28.8	34.6	51.4	59.6	65.6	68.2	66.6	58.5	48.8	38.2	30.6	
	5%	DB	24.1	26.9	37.1	58.3	74.3	79.6	80.2	76.1	67.0	53.9	37.7	28.8	
		MCWB	22.9	25.5	33.5	48.1	57.6	65.0	66.8	64.4	56.1	46.2	35.5	27.6	
	10%	DB	20.3	23.6	34.4	52.8	69.3	76.1	77.2	72.6	62.7	49.3	34.0	25.1	
		MCWB	19.2	22.1	31.4	44.4	55.1	63.2	65.9	62.5	53.6	43.9	32.4	23.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.8	32.2	38.3	55.7	64.3	70.6	72.5	70.5	63.4	53.3	42.5	32.8
			MCDB	32.0	33.7	43.3	67.3	79.6	81.0	83.1	79.7	73.2	63.0	43.9	33.8
2%		WB	26.4	28.9	35.6	52.3	61.2	68.1	70.4	68.0	59.8	50.2	38.5	30.7	
		MCDB	27.7	30.4	39.0	62.9	75.2	79.2	79.6	77.5	69.7	58.2	40.5	31.9	
5%		WB	22.9	25.6	33.7	48.5	58.5	66.2	68.7	65.7	57.0	46.9	35.6	27.6	
		MCDB	24.0	26.8	36.7	57.8	71.8	76.5	77.6	73.5	65.4	53.0	37.5	28.8	
10%		WB	19.3	22.1	31.6	44.9	56.0	64.4	66.9	63.5	54.5	44.1	32.7	23.9	
		MCDB	20.2	23.5	34.5	52.1	67.2	74.2	75.0	70.9	61.3	49.0	34.0	25.1	
Mean Daily Temperature Range		5% DB	MDBR	16.3	17.5	18.1	17.7	21.0	21.1	19.6	20.2	19.0	14.2	12.2	14.6
			MCDBR	15.7	13.3	17.0	26.0	29.6	27.2	25.1	26.2	27.6	24.2	11.2	13.1
	MCWBR		15.5	12.6	13.2	15.4	13.8	13.0	12.0	14.4	15.5	14.8	9.6	12.8	
	5% WB	MCDBR	16.1	12.7	14.6	25.1	27.0	24.4	22.0	23.2	24.4	21.6	11.2	12.8	
		MCWBR	16.0	12.2	11.9	15.5	13.9	12.5	11.6	13.9	14.5	13.9	9.9	12.7	
Clear-Sky Solar Irradiance	taub	0.286	0.318	0.371	0.408	0.404	0.405	0.438	0.411	0.366	0.349	0.315	0.287		
	taud	2.018	1.989	1.919	2.071	2.225	2.295	2.197	2.255	2.334	2.237	2.071	1.962		
	Ebn at Noon	184	225	241	254	265	266	253	252	249	217	169	145		
	Edh at Noon	21	32	45	45	41	39	42	37	30	25	20	17		
All-Sky Solar Radiation	RadAvg	275	600	1075	1459	1690	1867	1683	1367	935	470	227	172		
	RadStd	23	35	57	115	161	184	120	117	102	48	42	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page